

Business Continuity and Disaster Recovery Plan

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1. Purpose

Ensure continuity of Goya Ledger’s trust services (CA, TSA, OCSP, RA) during and after disruptive events. Fulfills ETSI TS 102 042 §7.4.9 and Chilean PSC accreditation requirements.

2. Service Tiers and Recovery Objectives

Service	RTO	RPO	Justification
OCSP Responder	15 min	0 (stateless)	Certificate validation must not stall
CRL Publication	1 hour	Last CRL	DS 181 requires timely revocation info
TSA	4 hours	Last token serial	Timestamp continuity, serial monotonicity
CA (issuance)	24 hours	Last issued cert	New certs can wait; revocation cannot
RA (proofing)	48 hours	Last proofing record	Human process, can queue
Audit Log	4 hours	0 (hash chain)	Regulatory requirement, tamper-evidence
API Gateway	15 min	N/A	Subscriber access to all services

RTO = Recovery Time Objective (max downtime). **RPO** = Recovery Point Objective (max data loss).

3. Architecture for Resilience

3.1 Data Protection

- **Blockchain state:** Replicated across consensus nodes (BFT/Raft)
- **Audit logs:** Append-only with SHA-256 hash chain; RocksDB WAL + checkpoints
- **CA keys:** M-of-N custodian shares (key ceremony); root key offline
- **Certificates/CRL:** Generated from CA key + serial state; reproducible
- **TSA serial counter:** Persisted; monotonicity verified on restart

3.2 Checkpoint and Snapshot

- `src/checkpoint.rs` — periodic state snapshots
- `src/storage/snapshot.rs` — RocksDB snapshot export/import
- `src/ordering/raft_storage.rs` — Raft log persistence
- Checkpoint frequency: every 1000 blocks or 1 hour (whichever first)

3.3 Geographic Distribution

Site	Role	Services
Primary (Santiago)	Active	All services
Secondary (backup)	Standby	OCSP, CRL mirror, read-only API
Offline vault	Cold storage	Root CA key shares, backup media

4. Disaster Scenarios

4.1 Single Node Failure

- **Detection:** Health check failure (< 30 seconds)
- **Response:** Automatic failover via Raft consensus
- **Recovery:** Replace node, restore from peer sync
- **RTO:** < 1 minute (consensus continues with remaining nodes)

4.2 Datacenter Outage

- **Detection:** All nodes unreachable from monitoring
- **Response:** Activate secondary site
- **Recovery:**
 1. Promote secondary to active
 2. Restore latest checkpoint to new primary
 3. Verify audit chain integrity
 4. Resume TSA with next serial number
 5. Publish fresh CRL
- **RTO:** 4 hours

4.3 CA Key Compromise

- See Incident Response Plan (GOYA-IRP-001) §5.3
- **Recovery:**
 1. Revoke compromised CA certificate
 2. Reconstruct CA key from custodian shares (M-of-N)
 3. Issue new intermediate CA certificate from root
 4. Re-issue all subscriber certificates
 5. Publish emergency CRL
- **RTO:** 24 hours

4.4 Data Corruption

- **Detection:** Audit chain verification failure (`verify_audit_chain()`)
- **Response:** Isolate affected storage
- **Recovery:**
 1. Identify last valid checkpoint before corruption
 2. Restore from checkpoint
 3. Replay transactions from peer nodes
 4. Verify chain integrity post-restore
- **RTO:** 4 hours
- **RPO:** Last valid checkpoint

4.5 Ransomware / Complete Data Loss

- **Recovery:**
 1. Rebuild from clean OS image
 2. Restore CA from offline key ceremony shares
 3. Restore latest off-site backup
 4. Sync blockchain state from peer network
 5. Verify all service integrity
- **RTO:** 24–48 hours

5. Backup Strategy

Data	Method	Frequency	Retention	Location
RocksDB state	Checkpoint snapshot	Hourly	30 days	Primary + secondary
Audit logs	Export + hash verification	Daily	7 years	Off-site encrypted
CA key shares	Key ceremony custodian	On generation	Permanent	Separate secure facilities
Configuration	Git repository	On change	Permanent	Remote repository
TSA serial state		Per token	7 years	With audit logs

Data	Method	Frequency	Retention	Location
	Atomic counter persist			

6. Recovery Procedures

6.1 Service Restart Order

1. Storage layer (RocksDB restore)
2. Consensus engine (Raft state recovery)
3. Audit log (verify hash chain)
4. CA (load intermediate key)
5. CRL (publish current list)
6. OCSP (start responder)
7. TSA (verify serial monotonicity)
8. RA (restore proofing queue)
9. API gateway (enable endpoints)

6.2 Verification Checklist

- ☐ `verify_audit_chain()` passes on restored log
- ☐ TSA serial number > last known serial
- ☐ NTP sync validated (`NtpTimeSource::validate()`)
- ☐ CRL published and accessible
- ☐ OCSP responds correctly for known serials
- ☐ CA can issue test certificate
- ☐ All API endpoints return 200 on health check

7. Testing

Test	Frequency	Scope
Checkpoint restore	Monthly	Restore from snapshot, verify state
Failover drill	Quarterly	Simulate primary failure, activate secondary
Full DR exercise	Annually	Complete rebuild from backups

Test	Frequency	Scope
Audit chain verification	Weekly	Run <code>verify_audit_chain()</code> on production log
CA key reconstruction	Annually	M-of-N share reassembly (test environment)

8. Regulatory References

- ETSI TS 102 042 §7.4.9 — Business continuity and disaster recovery
- Ley 19.799 Art. 17 — PSC obligations for service availability
- DS 181/2002 — Reglamento
- ISO 22301:2019 — Business continuity management systems

9. Document Control

Version	Date	Author	Changes
1.0	2026-08-05	Operations Lead	Initial draft